

Anant Mathur

PhD in Mathematics and Statistics, School of Mathematics and Statistics
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School of Mathematics and
Statistics, UNSW Sydney,
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A. Education

UNSW Sydney, Sydney, Australia 09/20 – 06/25
PhD in Statistics
Focus Area: Computational Statistics and Operations Research
Dissertation title: [Efficient Algorithms for Sparse Statistical Learning](#)
Supervisors: Dr Zdravko Botev, Dr Sarat Moka

UNSW Sydney, Sydney, Australia 01/15 - 12/19
Bachelor of Science (Honours), Statistics, 2019

B. Research areas

Methodological areas: Model compression, Optimization, Numerical analysis, Computational Statistics
Specific topics: Penalized optimization, Matrix computations, Monte-Carlo methods, Neural network pruning
Applications in genomics, quantitative risk management, and deep learning
Programming languages: Julia, Python, MATLAB, R

C. Awards and Honours

PhD thesis evaluated in the top 10% by both external examiners (UCLA, NCSU) and nominated for the UNSW Dean's Award for Outstanding PhD Theses (awaiting outcome)

Outstanding Statistics Talk, UNSW School of Mathematics and Statistics, Postgraduate Conference (2023)

UNSW Nominee for [JB Douglas Award](#) (2023)

D. Academic Service

▷ **Service as reviewer for journals**

Pattern recognition

▷ **Service as organizer/chair of technical sessions in conferences**

Probability and Optimization for Statistical Learning, INFORMS International Meeting, Singapore, 07/25

▷ **Honours students for whom significant thesis research guidance provided**

Ruiqi Dao, 01/25 – 10/25 (Primary advisor: Sarat Moka)

Seungjoo Lee, 01/24 – 10/24 (Primary advisor: Sarat Moka)

E. Research publications & preprints

▷ Manuscripts in preparation

[A. Mathur](#), S. Moka, B. Lique "Subset Selection in Generalized Linear Models via Continuous Optimization".

▷ Manuscripts under peer-review

[A. Mathur](#), S. Moka, Z. Botev "A Fast MM Algorithm for Group Lasso" [[PDF](#)].

▷ In peer-reviewed journals

[A. Mathur](#), S. Moka, Z. Botev, (2022). "Coordinate Descent for Variance-Component Models" *Algorithms*, 2022, 15 (10): 354.

▷ Papers in peer-reviewed conference proceedings

[A. Mathur](#), S. Moka, B. Lique, Z. Botev, "Group COMBSS: Group Selection via Continuous Optimisation", 2024 Winter Simulation Conference (WSC), In Proceedings of the 2024 *Winter Simulation Conference*, IEEE press, 2024, pp. 3217-3228

[A. Mathur](#), S. Moka, Z. Botev, "Feature Selection in Generalised Linear Models via the Lasso: To Scale or Not to Scale?", *15th OPT-ML Workshop, NeurIPS 2023*.

[A. Mathur](#), S. Moka, Z. Botev, "Column Subset Selection and Nyström Approximation via Continuous Optimisation", 2023 Winter Simulation Conference (WSC), In Proceedings of the 2023 *Winter Simulation Conference*, IEEE press, 2023. pp. 3601-3612.

[A. Mathur](#), S. Moka, Z. Botev, "Variance Reduction for Matrix Computations with Applications to Gaussian Processes", *VALUETOOLS 2021*, LNCS 13116, pp. 275-290.

F. Research talks

▷ Invited talks in Specialized Workshops/Conferences

SSA NSW Branch 2024 Annual Event

▷ Seminar Presentations

Macquarie University, School of Mathematical and Physical Sciences, 09/24

UNSW Sydney, School of Mathematical and Statistics, Statistics Seminar, 09/24

▷ Contributed talks in major conferences

INFORMS International Meeting, Singapore, 07/25

2023 Winter Simulation Conference, San Antonio, 12/23

25th International Congress on Modelling and Simulation (MODSIM 2023), Darwin, 07/23

Australian Mathematical Society Annual Meeting 2022, Sydney, 12/22

VALUETOOLS 2021 (Online), 10/2